PROJECT REVISION / 2026–2027

SmartCricket AI

Presentation Guide and Live Demonstration

An Explainable Multimodal Cricket Shot Analysis and Intelligent Virtual Coaching System

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Revised 5 November 2026

What this revision delivers

A broader feature demonstration with explicit synthetic-data labels, a documented practice-goal workflow, and an evidence-based account of the saved model results.

Accuracy status

63.93% is the saved local fusion Top-1 result. It was not rerun for this revision. Synthetic examples do not increase or establish classifier accuracy. The base paper reports 89% under its own setup.

College and guide details remain to be supplied by the student before submission.

Opening explanation

My project extends a published cricket-shot classification study into an application for capture, explanation, practice and tracking. The saved local fusion model has 63.93% Top-1 accuracy, compared with 56.17% for the local video baseline. I am not claiming to beat the paper's 89%. The new demonstration data is clearly labeled synthetic and is not part of that evaluation.

Suggested live demonstration

- Sign in as a player and show the dashboard's next step and existing photo/video report.
- Open Practice goals and explain the baseline, retest, shared joints and capture checks.
- Open Demo showcase and read the synthetic-data notice before presenting any values.
- Show the closer-target scenario, then the farther-target, camera-mismatch and visibility cases.
- Show a recovery day with no planned drills and export the labeled CSV or JSON.
- Finish with the recorded model-results slide and the real-training plan.

Answers to likely review questions

Question	Answer
Did accuracy increase in this revision?	No new training was run. The saved local comparison is +7.76 points; this revision adds features and fixes future model selection.
Why use dummy data?	It lets reviewers inspect feature behavior and edge cases without pretending that fictional activity is a real study.
Is 89.60% the project accuracy?	It is the saved fusion Top-3 score. Top-1 is 63.93%.
Does a smaller knee angle mean better batting?	No. We show a signed change and distance to a player-selected target, with capture limitations.
Is calibration a correctness guarantee?	No. Recorded aggregate ECE decreased; individual and out-of-domain predictions can still be wrong.
What remains unresolved?	Dataset count differs by five clips; paired-test column ordering needs audit; raw manifests and predictions are needed for reproduction.
How will real accuracy improve?	Use genuine labeled clips, fixed leakage-aware splits, training-only augmentation, validation-based selection and a held-out evaluation.

Slide-by-slide speaking notes

Slide 1: SmartCricket AI

Introduce the project as analysis and practice support. State that this revision adds features and honest demo coverage; it does not introduce a new trained accuracy score.

Slide 2: Abstract: The Project in One Slide

Give the problem, the paper starting point, the application contribution, then the saved local result. Use ‘recorded’ rather than implying a fresh experiment.

Slide 3: Why This Project

Explain the practical gap between a research classifier and a daily player workflow: capture, understandable results, practice context and saved follow-up.

Slide 4: Project Objectives

Separate completed application modules from the limited local model reproduction. The full preprocessing/training setup of the paper was not reproduced here.

Slide 5: Base paper and project contribution

Attribute the 89% result to the supplied paper. Explain that the local setup differs and that wider application scope is a separate contribution.

Slide 6: Existing System: CricShotNet (Base Paper)

Walk through cropping, frame sampling, segmentation, frame features and the GRU. These are the paper pipeline and authors' results, not measurements generated in this revision.

Slide 7: Existing System: Disadvantages

Describe scope gaps relative to this application. A temporal video model still observes movement; the missing component is explicit pose measurement, not all motion information.

Slide 8: Proposed System: Two Ways to Analyse a Batter

Contrast photo posture with video classification. Show how measurements and prediction outputs lead to reports and practice support. Mention optional live-frame uploads.

Slide 9: All Features at a Glance

Group features around capture, explanation, practice and tracking. Point reviewers to the new goals and synthetic showcase slides for the expanded workflow.

Slide 10: Proposed System: Advantages

Discuss useful inspection aids and low-confidence warnings without implying model certainty. Privacy depends on whether optional recognition or saving is enabled.

Slide 11: System Architecture

Follow a request from browser through authenticated API, preprocessing, analysis, storage and report generation. Installed classifiers are legacy models; newer training paths have no fresh evaluation here.

Slide 12: Software Description

Name the stack and each role. The optional LLM provider is configurable. Browser/device support and response time depend on the environment.

Slide 13: M1 User Interface and Sign-in

Show role-specific navigation and the player's next step. Explain that the historical screenshot is illustrative; current demo screenshots are labeled separately.

Slide 14: M2 Photo Posture Report

A landmark is an estimate. Low-visibility joints are omitted. One photograph supports a posture report, not recognition of an entire swing.

Slide 15: M3 and M4 Video Shot Analysis

Distinguish Top-1 prediction, Top-3 candidates and calibrated confidence. Heatmaps and attention support inspection and can be unavailable in movement-only mode.

Slide 16: M5 Compare with a Reference Clip

The administrator supplies the reference. DTW aligns sequences in time, but a difference from that reference is not automatically an error in technique.

Slide 17: M6 and M7 AI Coach

Describe stored-fact prompts, the numeric guard and fallback. This constrains some unsupported numbers without guaranteeing that every statement is true.

Slide 18: M9 Daily Training Coach

Explain check-in, recovery, planned drills, quiz and activity records. Levels and streaks describe engagement, not measured batting skill.

Slide 19: Live Mirror and Phone Mode

Pose estimation happens locally. Optional classifier requests send frames and saving a best moment uploads one. Physical webcam/microphone behavior was not revalidated here.

Slide 20: M8 History, Reports and Admin

Demonstrate retrieval and export. Administrators manage model evaluations and references; player-owned captures and goals enforce server-side ownership.

Slide 21: Recorded results: does pose help locally?

Explain Top-1 versus Top-3 and the local baseline gain. Attribute all figures to saved artifacts; do not claim the experiment was rerun or that it beats the paper.

Slide 22: Results: Trust, Robustness and Speed

Lower aggregate ECE is useful but not a per-prediction guarantee. Historical timings are not current universal benchmarks; refer to the new functional verification record.

Slide 23: Practice goals: baseline → retest

Walk through baseline, personal focus, practice, comparable retest, explicit save and PDF. The player selects the target and video frames; the app does not infer an ideal batting angle.

Slide 24: New: a clearly labeled demo showcase

State explicitly that the displayed practice values are synthetic. Demonstrate the behavior and interpretation rather than presenting fictional outcomes as measured improvement.

Slide 25: Demo showcase in the application

State explicitly that the displayed practice values are synthetic. Demonstrate the behavior and interpretation rather than presenting fictional outcomes as measured improvement.

Slide 26: How to read a demo comparison

State explicitly that the displayed practice values are synthetic. Demonstrate the behavior and interpretation rather than presenting fictional outcomes as measured improvement.

Slide 27: Evidence limits to resolve

Describe the count and statistical-ordering issues openly. Explain what raw manifests and paired predictions would resolve.

Slide 28: A valid path to higher real accuracy

Describe genuine labels, leakage-aware fixed splits, training-only augmentation and validation-based selection. A stronger training path remains an experiment until measured on held-out data.

Slide 29: Conclusion and Future Work

Close with the implemented feature contribution and saved local comparison. Keep real accuracy improvement as future measured work.

Slide 30: References

Identify the supplied base paper and local result artifacts as the direct sources for the headline numerical claims. Other literature is historical context from the original deck.

Slide 31: Questions and live demo

Explain the implemented capability and keep capture measurements, practice activity and research accuracy distinct.